

This document serves as your **Final API Design (DT-2)**. It acts as the contract between your Backend (the Kitchen) and your Frontend (the Table).

All endpoints follow the **REST** standard. They expect and return **JSON**. All data is automatically filtered by the `tenant_id` and `site_id` found inside the user's login token (JWT).

DT-2 — API DESIGN (FINAL)

Document Processing Analytics Platform

Base URL: `/api/v1`
Security: Authorization: Bearer `<JWT_TOKEN>` required for all except `/auth/login`.

1. Identity & Access (FR-5)

These endpoints handle logging in and selecting which physical location (Site) to view.

POST `/auth/login`

- **Description:** Authenticates user and returns a token.
- **Body:** `{ "email": "user@coca-cola.com", "password": "password123" }`
- **Response:**

```
{
  "token": "eyJhbGciOi...",
  "user": { "id": "uuid", "name": "John Doe", "role": "Admin" },
  "available_sites": [
    { "id": "uuid", "name": "Atlanta Plant" },
    { "id": "uuid", "name": "NY Distribution" }
  ]
}
```

2. Dashboard Endpoints (FR-1)

Used by the main landing page. Targets Layer 2 (Transactions) and Layer 3 (Error Catalog).

GET `/dashboard/summary` (FR-1.1)

- **Description:** Returns the 4 main counters. Uses the pre-aggregated columns in `TRANSACTIONS`.
- **Response:**

```
{
  "queued": 45,
  "processing": 12,
  "completed": 1250,
  "failed": 8
}
```

GET `/dashboard/throughput` (FR-1.2)

- **Description:** Number of completed files per day for the chart.
- **Params:** `days` (default 7).
- **Response:** `[ { "date": "2023-10-01", "count": 450 }, { "date": "2023-10-02", "count": 480 } ]`

GET `/dashboard/recent-failures` (FR-1.4)

- **Description:** Top 5 latest failed files to show on the dashboard table.
- **Response:**

```
{
  "data": [
    {
      "file_id": "uuid",
      "file_name": "invoice_99.pdf",
      "failed_at": "2023-10-21T14:00:00Z",
      "step": "Transform",
      "error_message": "Invalid Currency Code"
    }
  ]
}
```

3. Transaction Explorer (FR-2)

Allows browsing groups of files (Batches). Targets Layer 2.

GET /transactions (FR-2.1, 2.2, 2.3)

- **Description:** Searchable, filterable list of all batches.
- **Params:** `page`, `pageSize`, `status`, `search` (Batch ID), `startDate`, `endDate`.
- **Response:**

```
{
  "total_records": 500,
  "data": [
    {
      "id": "T-100",
      "status": "Completed",
      "total_files": 50,
      "submitted_at": "...",
      "completed_at": "..."
    }
  ]
}
```

GET /transactions/{id}/files (FR-2.4)

- **Description:** Lists all individual files belonging to one specific batch.
- **Response:** [{ "id": "uuid", "name": "doc1.pdf", "status": "Success", "current_step": "Published" }]

4. File Detail & Error Analysis (FR-3)

Deep dive into why a file failed. Targets Layer 2 (History) & Layer 3 (Catalog).

GET /files/{id}/history (FR-2.5)

- **Description:** The "Timeline" view of every step a file took.
- **Response:**

```
[
  { "step": "Upload", "status": "Success", "timestamp": "...", },
  { "step": "Validate", "status": "Failed", "error_code": "ERR_VAL_01" }
]
```

GET /errors/analytics (FR-3.1, 3.2)

- **Description:** Data for the Error Analysis page (Top 10 errors and 30-day trend).
- **Response:**

```
{
  "top_errors": [ { "code": "EXTRACT_FAIL", "count": 150, "fix": "Check OCR" } ],
  "trend": [ { "date": "...", "count": 5 } ]
}
```

5. Activity Log & Extraction (FR-4)

Targets *Layer 3 (Audit)* and *Layer 4 (Fields)*.

GET /activity-log (FR-4)

- **Params:** `entity_id` (optional), `page`.
- **Response:** [{ "timestamp": "...", "event": "Batch Submitted", "user": "System" }]

GET /files/{id}/extracted-fields (Layer 4)

- **Description:** Shows the key-value data found inside the document.
- **Response:** [{ "field": "TotalAmount", "value": "150.00", "confidence": 0.95 }]

Implementation Rules (The "How-To")

1. The "Envelope" Rule

All list responses must follow this format so the Frontend knows how to build the "Pagination" buttons:

```
{
  "data": [],
  "meta": {
    "total_count": 1000,
    "current_page": 1,
    "page_size": 50,
    "total_pages": 20
  }
}
```

2. Error Handling (Standard Status Codes)

- **401 Unauthorized:** User didn't provide a token or token expired.
- **403 Forbidden:** User is logged in but trying to see a `site_id` they don't own (Tenant Isolation failure).
- **404 Not Found:** The Batch ID or File ID doesn't exist.
- **500 Internal Error:** The database is down or code crashed.

3. Tenant Isolation (Backend Logic)

When the Frontend calls `GET /api/v1/transactions`, the Backend code **MUST NOT** trust the user to provide a `tenant_id`.

- **Step 1:** Read the JWT.
 - **Step 2:** Extract `tenant_id = 'coca-cola'`.
 - **Step 3:** Perform SQL: `SELECT * FROM transactions WHERE tenant_id = 'coca-cola'`.
-